Important notice – limits of this FTO pre-screen

1. This is an automated (AI-assisted) pre-screen. It is not legal advice and not a freedom-to-operate opinion, and it does not replace review by a patent attorney.
2. The screened corpus is restricted to the LNP/lipid field (EP/US/WO; defined CPC classes, keywords and applicants). Rights outside this restriction – including utility models, national rights of other countries and supplementary protection certificates – are not covered.
3. Markush/generic claims may cover structures that are not individually indexed in the corpus. No Markush search (e.g. CAS STN MARPAT) was performed. The formula test is only applied to claims whose text is in the corpus; formulae embedded as images may be missing.
4. Legal status (grant, opposition, limitation, lapse, term, PTA/PTE, SPC, UPC opt-out) reflects a data snapshot of 2026-08-23 and may be stale or incomplete. Expiry dates are estimates.
5. Claim texts may be machine translations or OCR text; only the text in the language of proceedings, as granted or maintained, is authoritative.
6. Pending applications: claims may change before grant; divisionals and continuations may be filed. Applications not yet published (up to 18 months from priority) are not covered.
7. The assessment rests on the product/process data and assumptions listed under "Subject of the screen". If they change, the result changes.
8. Claim construction, equivalence and exemptions (experimental use, Bolar, safe harbor) depend on the individual case and current case law; prosecution files were not reviewed.
9. AI models make mistakes (wrong substituent mapping, wrong chain-length counts, overlooked provisos). Verify quotes and feature mappings against the original document before any decision.

LNP Patent Desk

FREEDOM-TO-OPERATE PRE-SCREENING

FTO pre-screen (not a legal opinion): Testprodukt LNP-1

Pre-screening, not legal advice

CREATED
2026-11-10 12:00

DEPTH
standard

MARKETS
DE, EP, US

CORPUS SNAPSHOT
EP full text 2026-11-10; US full text 2026-11-10; EP Register 2026-08-23; SureChEMBL 2026-11-03

ID
op17f3cf07ae2d4452

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1. Summary

OVERALL RISK

very high (5)

●●●●●

Highest level across the target markets, computed by the app from the individual assessments. Driven by EP1000001B1, EP1000008B1, EP1000009B1.

DE	●●●●● very high (5)
EP / UPC	●●●●● very high (5)
US	●●●●○ high (4)
Pending applications	●●●○○ medium (3)

7 patent rights analysed · 4 not analysed

Overall risk per market (computed by the app from the individual assessments)

MARKET	RISK	DRIVING PATENT RIGHTS
DE	very high (5)	EP1000001B1, EP1000008B1, EP1000009B1
EP / UPC	very high (5)	EP1000001B1, EP1000008B1, EP1000009B1
US	high (4)	US9000003B2

Pending applications (separate; claims may change): medium (3) - US20250000006A1

Risk matrix

PATENT RIGHT	MARKET	RELEVANCE	RISK	CLAIM CHART / RED TEAM	LITERAL INFRINGEMENT (WORST CLAIM)	LEGAL STATUS	EXPIRY (ESTIMATE)	NOTES
EP1000001B1	DE/EP	A	very high (5)	very high (5) high (4)	literally met	granted, in force in DE (snapshot 2026-08-23)	2035-06-01 (estimate)	literal result recomputed, dependent claims: quote not verified, dependent claims: chain result recomputed
EP1000008B1	DE/EP	A	very high (5)	very high (5) / -	literally met	unknown, DE part unknown (unverified, no record; treated as in force)	2035-06-01 (estimate)	
EP1000009B1	DE/EP	A	very high (5)	very high (5) high (4)	literally possible	granted, in force in DE (snapshot 2026-08-23)	2035-06-01 (estimate)	text shortened, literal result recomputed
US9000003B2	US	B	high (4)	low (2) high (4)	literally possible	granted, maintenance fees not checked (snapshot 2026-11-10)	2035-11-01 (estimate)	escalated by the red-team review
EP1000004B1	DE/EP	A	? cannot be assessed	? cannot be assessed very low (1)	literally possible	granted, in force in DE (snapshot 2026-08-23)	2035-06-01 (estimate)	formula not available, literal result recomputed, verdict set to unclear by the app
EP1000002B1	DE/EP	A	medium (3)	medium (3) low (2)	literally possible	granted, not in force in DE, opposition/appeal pending (snapshot 2026-08-23)	2035-06-01 (estimate)	literal result recomputed, quote not verified, verdict set to unclear by the app
US20250000006A1	US	B	medium (3)	medium (3) medium (3)	literally possible	pending (snapshot 2026-11-10)	2035-06-01 (estimate)	

Summary assessment (Claude; from the individual assessments, not binding)

> Im geprüften Korpus bestehen mehrere Risiken.
>
> - Punkt eins
> - Punkt zwei <script>alert(9)</script>

App notes on the synthesis: overall risk differs from the computed one

App notes on this run

- Step B.EP1000009B1: input shortened (description excerpt d2 -101 bytes, description excerpt d1 -170 bytes, claim 1 -265448 bytes)
- Step E.EP1000009B1: input shortened (description excerpt d2 -101 bytes, description excerpt d1 -170 bytes, claim 1 -256483 bytes)

2. Subject of the screen (product profile as understood)

Testprodukt LNP-1

Ionizable lipid

Name	ALC-0315 / ALC-0315
SMILES	CCCCCCCCC(CCCCCC)C(=O)OCCCCCCN(CCCCO)CCCCCCOC(=O)C(CCCCCC)CCCCCCCC
InChIKey	QGWBEETXHOVFQS-UHFFFAOYSA-N
Formula (computed by the app (OpenChemLib), not by the model)	C48H95NO5
Molecular weight	766.3
Structure description	Tertiäres Amin mit zwei Esterarmen. [Computed by the app: head group hydroxyalkyl-tert-amine (OCCCCN); amine: tertiary; linkers: 2 x ester; chains: Acyl C16 branched at position 2 (8+6) | Acyl C16 branched at position 2 (8+6).]

Other lipids

ROLE	NAME	SMILES	PEG-MW
helper lipid / phospholipid	DSPC / 1,2-distearoyl-sn-glycero-3-phosphocholine	DSPC-SMILES	
sterol	cholesterol / Cholesterol	CHOL-SMILES	
PEG-lipid	ALC-0159 / ALC-0159	PEG-SMILES	2000

Composition (mol-% of total lipid)

ionizable lipid	46.3
phospholipid	9.4
sterol	42.7
PEG-lipid	1.6
Basis	total lipid, nominal (as charged)
N/P	6

Payload

Type	mRNA
Nucleoside modification	m1Ψ

Particle

Size (nm)	80
PDI	0.1

Process

Mixing principle	T-junction
Flow rate ratio aqueous:organic	3
Ethanol at mixing (%)	25

Use

Route of administration	i.m.

Acts and markets

Development stage preclinical

Markets DE, EP, US

COUNTRY	ACT	TIMING	PLANNED FROM	PURPOSE
DE	making	current		

Product feature list

ID	CATEGORY	STATEMENT	VALUE	SOURCE	CERTAINTY
P1	ratio	Ionisierbares Lipid 46,3 mol-%	46.3 mol-%	user	high
P2	lipid structure	Formula / molecular weight: C48H95NO5 / 766.3	C48H95NO5	computed	high
P3	lipid structure	Head group and amine: hydroxyalkyl-tert-amine, tertiary (1)	hydroxyalkyl-tert-amine	computed	high
P4	lipid structure	Linkers: 2 x ester	2 x ester	computed	high
P5	lipid structure	Chain 1: Acyl C16 branched at position 2 (8+6) (acyl: carbonyl carbon counted)	16 C	computed	high
P6	lipid structure	Chain 2: Acyl C16 branched at position 2 (8+6) (acyl: carbonyl carbon counted)	16 C	computed	high
P7	lipid structure	Total carbon atoms: 48	48 C	computed	high

Open questions on the product

- Gemessen oder nominal?

App notes on the profile

- SMILES from the model reply discarded (differs from the input).

Description by Claude

> Gegenstand.

3. Search record (for reproducibility)

Corpus and snapshots surechembl 2026-11-03; ep_register 2026-08-23; ep_text 2026-11-10; us_text 2026-11-10; corpus T

Size patents 12

Candidates 11

Triage A 5, B 2, C 1

Analysed 7

Structure search (exact / substructure / similarity per lipid)

LIPID	ROLE	MODE	COMPOUNDS	OCCURRENCES
ALC-0315	ionizable	exact	1	2
ALC-0315	ionizable	substructure	0	0

LIPID	ROLE	MODE	COMPOUNDS	OCCURRENCES
ALC-0315	ionizable	similarity	0	0
DSPC	helper	exact	0	0
cholesterol	sterol	exact	0	0
ALC-0159	peg	exact	0	0
ALC-0159	peg	substructure	0	0
ALC-0159	peg	similarity	0	0

Text search (generated boolean queries, verbatim)

	QUERY	HITS	EVALUATED
Names/codes in the claims	claims:("ALC-0315" OR "ALC-0159")	8	8
Names/codes in title, abstract, claims	("ALC-0315" OR "ALC-0159")	8	8
LNP with cationic/ionizable lipid (independent claims)	same:(("lipid nanoparticle" OR "lipid nanoparticles" OR "nucleic acid-lipid particle" OR "lipid particle" OR "lipid particles" OR lipidnanopartikel OR "lipid-nanopartikel" OR "lipidpartikel" OR liposom* OR "lipid vesicle") AND ("cationic lipid" OR "ionizable lipid" OR "ionisable lipid" OR "amino lipid" OR "kationisches lipid" OR "kationische lipid" OR "ionisierbares lipid" OR "ionisierbare lipid")) indep:1	8	8
Molar ratios / mol-%	same:(("cationic lipid" OR "ionizable lipid" OR "ionisable lipid" OR "amino lipid" OR "kationisches lipid" OR "kationische lipid" OR "ionisierbares lipid" OR "ionisierbare lipid") AND ("mol %" OR "mol%" OR "mol percent" OR "molar ratio" OR molprozent OR "mol-%" OR "molverhältnis") AND (cholesterol OR cholesterin OR sterol OR phospholipid OR "non-cationic lipid" OR "neutral lipid")) indep:1	8	8
PEG-/polymer-conjugated lipid	same:(("conjugated lipid" OR "peg-lipid" OR "pegylated lipid" OR "peg lipid" OR "polymer conjugated lipid" OR "polyethylene glycol" OR "pegyliertes lipid" OR "polyethylenglykol") AND ("lipid nanoparticle" OR "lipid nanoparticles" OR "nucleic acid-lipid particle" OR "lipid particle" OR "lipid particles" OR lipidnanopartikel OR "lipid-nanopartikel" OR "lipidpartikel" OR liposom* OR "lipid vesicle")) indep:1	8	8
Lipid compound claims with a generic formula (C07C/C07D)	indep:("compound of formula" OR "lipid of formula" OR "compound having" OR "having the structure" OR "following structure" OR "verbindung der formel" OR "lipid der formel" OR "compound of the formula") cpc:C07C,C07D	8	8
Payload mRNA in LNP	same:((mrna OR "messenger rna" OR "messenger ribonucleic acid" OR "ribonucleic acid" OR rna OR boten*) AND ("lipid nanoparticle" OR "lipid nanoparticles" OR "nucleic acid-lipid particle" OR "lipid particle" OR "lipid particles" OR lipidnanopartikel OR "lipid-nanopartikel" OR "lipidpartikel" OR liposom* OR "lipid vesicle") AND ("cationic lipid" OR "ionizable lipid" OR "ionisable lipid" OR "amino lipid" OR "kationisches lipid" OR "kationische lipid" OR "ionisierbares lipid" OR "ionisierbare lipid")) indep:1	8	8
Manufacturing process (mixing, ethanol, microfluidics)	same:((mixing OR mischen OR vermischen OR "t-connector" OR "t-junction" OR microfluidic* OR mikrofluid* OR herringbone OR "impinging jet") AND (ethanol OR alcohol OR alkohol OR "organic solvent") AND lipid* AND (nanoparticle* OR particle* OR vesicle* OR liposom* OR partikel* OR nanopartikel*)) indep:1	8	8
Particle parameters (size, PDI)	same:((diameter OR durchmesser OR "particle size" OR polydispersity OR polydispersität* OR "mean size") AND ("lipid nanoparticle" OR "lipid nanoparticles" OR "nucleic acid-lipid particle" OR "lipid particle" OR "lipid particles" OR lipidnanopartikel OR "lipid-nanopartikel" OR "lipidpartikel" OR liposom* OR "lipid vesicle") AND ("cationic lipid" OR "ionizable lipid" OR "ionisable lipid" OR "amino lipid" OR "kationisches lipid" OR "kationische lipid" OR "ionisierbares lipid" OR "ionisierbare lipid")) indep:1	8	8

Key families of the watched applicants

TIER	HITS
1	1
2	2

- 1 WO publication(s) among the hits: not enforceable, used only as pointers to the EP/US family members.

Description by Claude

> Recherche.

4. Individual assessment of the analysed patent rights

EP1000001B1 - TITLE EP1000001B1

●●●●● very high (5)

Proprietor/applicant	Assignee 0
Assessed for	DE, EP (EP)
Triage	A: Grund EP1000001B1
Found via	Key family (tier 1) of a watched applicant: ARB-069 (Arbutus/Genevant/Protiva) | Text search "Names/codes in the claims" (claim 1) | Text search "Names/codes in title, abstract, claims" (claim 1) | Text search "LNP with cationic/ionizable lipid (independent claims)" (claim 1)
Legal status	granted, in force in DE (snapshot 2026-08-23)
Expiry (estimate)	2035-06-01 (estimate)
Basis	filing date 2015-06-01 + 20 years (Art. 63 EPC); SPC, surrender and renewal fees after the snapshot date not taken into account
Risk (final = the more adverse of chart and red team)	very high (5) [Claim chart / red team: very high (5) / high (4)]
Risk according to the model	medium (3)
Flags	literal result recomputed, dependent claims: quote not verified, dependent claims: chain result recomputed
Claim language	en

App checks

- Claim 1: result recomputed from the feature rows (literally possible → literally met)
- Risk raised from medium (3) to very high (5): an independent claim is literally met according to the feature rows
- Claim 1: red-team result more favourable (literally possible), not adopted
- The red-team review considers a lower risk arguable (high (4)): not adopted automatically, question for counsel

Short verdict (Claude)

Kurzfazit EP1000001B1

> Text EP1000001B1

Claim chart: Claim 1 (composition)

Result (computed from the feature rows)	literally met
Model	literally possible
Red team	literally possible
Final	literally met
Flags	literal result recomputed

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*A nucleic acid-lipid particle comprising*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M1 — Construction: broad = weit; narrow =
M2	*a cationic lipid comprising from 40 mol % to 65 mol % of the total lipid present in the particle*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M2 — Construction: broad = umfasst ionisierbare Lipide; narrow = nur permanent kationisch — Definition [d1]: "the term "cationic lipid" refers to any lipid that carries a net positive charge at a selected pH" — Numeric comparison: 40 - 65 mol %; Product 46.3; Δ 6.3
M3	*a non-cationic lipid comprising a mixture of a phospholipid and cholesterol*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M3 — Construction: broad = weit; narrow =
M4	*a conjugated lipid that inhibits aggregation of particles comprising from 0.5 mol % to 2 mol %*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M4 — Construction: broad = weit; narrow =

Dependent claims (fallback positions)

CLAIM	ADDED FEATURE (QUOTE)	VERDICT	CHAIN (APP)	NAMED INDIVIDUALLY
2	wherein the nucleic acid is an mRNA	met: mRNA	met	n/a
3	wherein the conjugated lipid is a PEG-lipid having a C12 anchor [NOT VERIFIED: quote is not in the claim text]	unclear: C14	possible	no

Fallback positions

- Claim 1 → Claim 2 (Anspruch 3): r

Construction hints from the dependent claims

- conjugated lipid (Claim 3): weiter als PEG-Lipid

Red team (patentee view)

Further routes of attack

- indirect infringement (medium): Lieferung

Revised risk of the red team: high (4). rt

Open questions for counsel

- Gemessene mol-% im Partikel?
- Analytik

Design-around ideas (ideas only: technical hypotheses, to be validated technically and legally)

- M2: mol-% unter 40 [narrow margin] - fachlich zu validieren

Exemptions and role (model view)

Experimental-use exemption (§ 11 No. 2 PatG)	open
Regulatory (Bolar) exemption (§ 11 No. 2b PatG)	yes
US safe harbor (35 U.S.C. § 271(e)(1))	n/a
Tipping point	ab Angebot Ausnahmen
Direct infringement by the user	yes
Indirect / contributory infringement	open Rolle

EP1000008B1 - TITLE EP1000008B1

●●●●● very high (5)

Proprietor/applicant	Assignee 7
Assessed for	DE, EP (EP)
Triage	A: Grund EP1000008B1
Found via	Text search "Names/codes in the claims" (claim 1) | Text search "Names/codes in title, abstract, claims" (claim 1) | Text search "LNP with cationic/ionizable lipid (independent claims)" (claim 1)
Legal status	unknown, DE part unknown (unverified, no record; treated as in force)
Expiry (estimate)	2035-06-01 (estimate)
Basis	computed by the app from the filing date 2015-06-01 + 20 years (no legal-status record); PTA/PTE/SPC not taken into account
Risk (final = the more adverse of chart and red team)	very high (5) [Claim chart / red team: very high (5) / -]
Risk according to the model	very high (5)
Claim language	en

Short verdict (Claude)

Kurzfazit EP1000008B1

> Text EP1000008B1

Claim chart: Claim 1 (composition)

Result (computed from the feature rows)	literally met
Model	literally met

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*A lipid nanoparticle comprising a cationic lipid*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M1 — Construction: broad = weit; narrow =
M2	*wherein the particle has a mean diameter of 60 nm to 120 nm*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M2 — Construction: broad = weit; narrow =

Open questions for counsel

- Gemessene mol-% im Partikel?

Design-around ideas (ideas only: technical hypotheses, to be validated technically and legally)

- M2: mol-% unter 40 [narrow margin] - fachlich zu validieren

Exemptions and role (model view)

Experimental-use exemption (§ 11 No. 2 PatG)	open
Regulatory (Bolar) exemption (§ 11 No. 2b PatG)	yes
US safe harbor (35 U.S.C. § 271(e)(1))	n/a
Tipping point	ab Angebot Ausnahmen
Direct infringement by the user	yes
Indirect / contributory infringement	open Rolle

EP1000009B1 - TITLE EP1000009B1

●●●●● very high (5)

Proprietor/applicant	Assignee 8
Assessed for	DE, EP (EP)
Triage	A: Grund EP1000009B1
Found via	Text search "Names/codes in the claims" (claim 1) | Text search "Names/codes in title, abstract, claims" (claim 1) | Text search "LNP with cationic/ionizable lipid (independent claims)" (claim 1)
Legal status	granted, in force in DE (snapshot 2026-08-23)
Expiry (estimate)	2035-06-01 (estimate)
Basis	filing date 2015-06-01 + 20 years (Art. 63 EPC); SPC, surrender and renewal fees after the snapshot date not taken into account
Risk (final = the more adverse of chart and red team)	very high (5) [Claim chart / red team: very high (5) / high (4)]
Risk according to the model	very high (5)
Flags	text shortened, literal result recomputed
Claim language	en

App checks

- Claim 1: text shortened, result at most "possible"
- Claim 1: result recomputed from the feature rows (literally met → literally possible)
- Claim 1: red-team result more favourable (literally not met), not adopted
- The red-team review considers a lower risk arguable (high (4)): not adopted automatically, question for counsel

Short verdict (Claude)

Kurzfazit EP1000009B1

> Text EP1000009B1

Claim chart: Claim 1 (composition)

Result (computed from the feature rows)	literally possible
Model	literally met
Red team	literally not met
Final	literally possible
Flags	text shortened, literal result recomputed

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*A lipid nanoparticle comprising a lipid selected from the group consisting of*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M1 — Construction: broad = weit; narrow =
M2	*and an mRNA.*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M2 — Construction: broad = weit; narrow =

Red team (patentee view)

Further routes of attack

- indirect infringement (medium): Lieferung

Revised risk of the red team: high (4). rt

Open questions for counsel

- Gemessene mol-% im Partikel?
- Analytik

Design-around ideas (ideas only: technical hypotheses, to be validated technically and legally)

- M2: mol-% unter 40 [narrow margin] - fachlich zu validieren

Exemptions and role (model view)

Experimental-use exemption (§ 11 No. 2 PatG)	open
Regulatory (Bolar) exemption (§ 11 No. 2b PatG)	yes
US safe harbor (35 U.S.C. § 271(e)(1))	n/a
Tipping point	ab Angebot Ausnahmen
Direct infringement by the user	yes
Indirect / contributory infringement	open Rolle

US9000003B2 - TITLE US9000003B2

●●●●○ high (4)

Proprietor/applicant	Assignee 2
Assessed for	US (US)
Triage	B: Grund US9000003B2
Found via	Text search "Names/codes in the claims" (claim 1) | Text search "Names/codes in title, abstract, claims" (claim 1) | Text search "LNP with cationic/ionizable lipid (independent claims)" (claim 1)
Legal status	granted, maintenance fees not checked (snapshot 2026-11-10)
Expiry (estimate)	2035-11-01 (estimate)
Basis	earliest effective US filing date 2015-06-01 + 20 years + PTA 153 days; terminal disclaimer, PTE and maintenance fees not checked
Risk (final = the more adverse of chart and red team)	high (4) [Claim chart / red team: low (2) / high (4)]
Risk according to the model	low (2)
Flags	escalated by the red-team review
Claim language	en

Short verdict (Claude)

Kurzfazit US9000003B2

> Text US9000003B2

Claim chart: Claim 1 (composition)

Result (computed from the feature rows)	literally not met
Model	literally not met
Red team	literally possible
Final	literally possible

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*A lipid nanoparticle comprising an ionizable lipid, a phospholipid, a sterol and a PEG-lipid*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M1 — Construction: broad = weit; narrow =
M2	*wherein the PEG-lipid is present at less than 1.0 mol %*	✔ found verbatim in the claim	1,6 mol-% [P1]	☐ not met	Begruendung M2 — Construction: broad = weit; narrow = — Red team (patentee view): finding is shaky

Claim chart: Claim 8 (composition)

Result (computed from the feature rows)	literally not met
Model	literally not met
Red team	literally not met
Final	literally not met

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*A pharmaceutical composition comprising the lipid nanoparticle and an mRNA encoding an antigen*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M1 — Construction: broad = weit; narrow =
M2	*wherein the composition is lyophilized*	✔ found verbatim in the claim	46,3 mol-% [P1]	☐ not met	Begruendung M2 — Construction: broad = weit; narrow =

Red team (patentee view)

CLAIM	FEATURE	ATTACK	PATENTEE ARGUMENT	COUNTER-ARGUMENT	OUTCOME
1	M2	product facts	Gemessene Zusammensetzung kann unter 1,0 mol-% liegen	Nominalwert 1,6	finding is shaky → unclear

Further routes of attack

- indirect infringement (medium): Lieferung

Revised risk of the red team: high (4). rt

Open questions for counsel

- Gemessene mol-% im Partikel?
- Analytik

Design-around ideas (ideas only: technical hypotheses, to be validated technically and legally)

- M2: mol-% unter 40 [narrow margin] - fachlich zu validieren

Exemptions and role (model view)

Experimental-use exemption (§ 11 No. 2 PatG)	open
Regulatory (Bolar) exemption (§ 11 No. 2b PatG)	yes
US safe harbor (35 U.S.C. § 271(e)(1))	n/a
Tipping point	ab Angebot
	Ausnahmen
Direct infringement by the user	yes
Indirect / contributory infringement	open
	Rolle

EP1000004B1 - TITLE EP1000004B1

? cannot be assessed

Proprietor/applicant	Assignee 3
Assessed for	DE, EP (EP)
Triage	A: Grund EP1000004B1
Found via	Possible generic formula claim: ALC-0315 is disclosed in the document and an independent claim contains a formula that exists only as an image – formula test needed | Exact structure: ALC-0315 = SCHEMBL1 in the claims (SureChEMBL occurrence, no evidence of claim scope) | Text search "Names/codes in the claims" (claim 1) | Text search "Names/codes in title, abstract, claims" (claim 1) | Text search "LNP with cationic/ionizable lipid (independent claims)" (claim 1)
Legal status	granted, in force in DE (snapshot 2026-08-23)
Expiry (estimate)	2035-06-01 (estimate)
Basis	filing date 2015-06-01 + 20 years (Art. 63 EPC); SPC, surrender and renewal fees after the snapshot date not taken into account
Risk (final = the more adverse of chart and red team)	cannot be assessed [Claim chart / red team: cannot be assessed / very low (1)]
Risk according to the model	very low (1)
Flags	formula not available, literal result recomputed, verdict set to unclear by the app
Claim language	en

App checks

- Claim 1: result recomputed from the feature rows (literally not met → literally possible)
- Risk raised from very low (1) to medium (3): no feature is clearly not met
- Risk raised from medium (3) to cannot be assessed: claim text incomplete: the formula exists only as an image and was not read
- Claim 1: red-team result more favourable (literally not met), not adopted
- The red-team review considers a lower risk arguable (very low (1)): not adopted automatically, question for counsel

Short verdict (Claude)

Kurzfazit EP1000004B1

> Text EP1000004B1

Claim chart: Claim 1 (compound)

Result (computed from the feature rows)	literally possible
Model	literally not met
Red team	literally not met
Final	literally possible
Flags	formula not available, literal result recomputed

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*A compound having the following structure of Formula (I): [chem]*	✔ found verbatim in the claim	46,3 mol-% [P1]	unclear (set to "unclear" by the app: formula not available; Model: not met)	Begruendung M1 — Construction: broad = weit; narrow =
M2	*G1 is C1-C6 alkylene*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M2 — Construction: broad = weit; narrow =

Formula test (Markush) M1: formula not available (Model: does not fall under the formula)

Scaffold	guessed
Distance (failing variables)	1
Confidence	medium
Named individually	no

VARIABLES	DEFINITION	QUOTE CHECK
R1	R1 and R2 are each independently C6-C24 alkyl	found verbatim in the claim

MAPPINGS M1 (ALL MET: NO)	FRAGMENT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
R1	C16	not met	x

Dependent claims (fallback positions)

CLAIM	ADDED FEATURE (QUOTE)	VERDICT	CHAIN (APP)	NAMED INDIVIDUALLY
2	A lipid nanoparticle comprising the compound of claim 1.	unclear: x	possible	n/a

Fallback positions

- Claim 1 → Claim 2 (Anspruch 3): r

Construction hints from the dependent claims

- conjugated lipid (Claim 3): weiter als PEG-Lipid

Red team (patentee view)

Further routes of attack

- indirect infringement (medium): Lieferung

Revised risk of the red team: very low (1). rt

Open questions for counsel

- Gemessene mol-% im Partikel?
- Analytik

Design-around ideas (ideas only: technical hypotheses, to be validated technically and legally)

- M2: mol-% unter 40 [narrow margin] - fachlich zu validieren

Exemptions and role (model view)

Experimental-use exemption (§ 11 No. 2 PatG)	open
Regulatory (Bolar) exemption (§ 11 No. 2b PatG)	yes
US safe harbor (35 U.S.C. § 271(e)(1))	n/a
Tipping point	ab Angebot Ausnahmen
Direct infringement by the user	yes
Indirect / contributory infringement	open Rolle

EP1000002B1 - TITLE EP1000002B1

●●●○○ medium (3)

Proprietor/applicant	Assignee 1
Assessed for	DE, EP (EP)
Triage	A: Grund EP1000002B1
Found via	Text search "Names/codes in the claims" (claim 1) | Text search "Names/codes in title, abstract, claims" (claim 1) | Text search "LNP with cationic/ionizable lipid (independent claims)" (claim 1)
Legal status	granted, not in force in DE, opposition/appeal pending (snapshot 2026-08-23); not in force in DE according to the register; possibly in force in other validation states
Expiry (estimate)	2035-06-01 (estimate)
Basis	filing date 2015-06-01 + 20 years (Art. 63 EPC); SPC, surrender and renewal fees after the snapshot date not taken into account
Risk (final = the more adverse of chart and red team)	medium (3) [Claim chart / red team: medium (3) / low (2)]
Risk according to the model	very low (1)
Flags	literal result recomputed, quote not verified, verdict set to unclear by the app
Claim language	en

App checks

- Claim 1: result recomputed from the feature rows (literally not met → literally possible)
- Risk raised from very low (1) to medium (3): no feature is clearly not met
- Claim 1: red-team result more favourable (literally not met), not adopted
- The red-team review considers a lower risk arguable (low (2)): not adopted automatically, question for counsel

Short verdict (Claude)

Kurzfazit EP1000002B1

> Text EP1000002B1

Claim chart: Claim 1 (manufacturing process)

Result (computed from the feature rows)	literally possible
Model	literally not met
Red team	literally not met
Final	literally possible
Flags	literal result recomputed

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*mixing an ethanolic lipid solution with an aqueous nucleic acid solution in a T-connector*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M1 — Construction: broad = weit; narrow =
M2	*wherein the mixing is performed at a Reynolds number above 2000 in a turbulent regime*	✖ NOT VERIFIED: quote is not in the claim text	46,3 mol-% [P1]	unclear (set to "unclear" by the app: quote not verified; Model: not met)	<img src=x onerror=alert(3)> **fett** x — Construction: broad = weit; narrow =
M3	*<script>alert(1)</script> <img src=x onerror=alert(2)>*	✔ found verbatim in the claim	46,3 mol-% [P1]	? unclear	Begruendung M3 — Construction: broad = weit; narrow = — Clarification: Messung
M4	*wherein the flow rate ratio is from 2:1 to 4:1.*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M4 — Construction: broad = weit; narrow =

Equivalence questions for counsel (NOT decided here)

- M2: Replaced means = laminar; model view, not a decision: low. eq

Red team (patentee view)

Further routes of attack

- indirect infringement (medium): Lieferung

Revised risk of the red team: low (2). rt

Open questions for counsel

- Gemessene mol-% im Partikel?
- Analytik
- Claim 1 M2: NOT VERIFIED: quote is not in the claim text - check against the original document
- Claim 1 M3: Messung
- Claim 1: equivalence for M2 (not decided)

Design-around ideas (ideas only: technical hypotheses, to be validated technically and legally)

- M2: mol-% unter 40 [narrow margin] - fachlich zu validieren

Exemptions and role (model view)

Experimental-use exemption (§ 11 No. 2 PatG)	open
Regulatory (Bolar) exemption (§ 11 No. 2b PatG)	yes
US safe harbor (35 U.S.C. § 271(e)(1))	n/a
Tipping point	ab Angebot
	Ausnahmen
Direct infringement by the user	yes
Indirect / contributory infringement	open
	Rolle

US20250000006A1 - TITLE US20250000006A1

●●●○○ medium (3)

Proprietor/applicant	Assignee 5
Assessed for	US (US)
Triage	B: Grund US20250000006A1
Found via	Text search "Names/codes in the claims" (claim 1) | Text search "Names/codes in title, abstract, claims" (claim 1) | Text search "LNP with cationic/ionizable lipid (independent claims)" (claim 1)
Legal status	pending (snapshot 2026-11-10)
Expiry (estimate)	2035-06-01 (estimate)
Basis	earliest effective US filing date 2015-06-01 + 20 years; terminal disclaimer, PTE and maintenance fees not checked
Risk (final = the more adverse of chart and red team)	medium (3) [Claim chart / red team: medium (3) / medium (3)]
Risk according to the model	medium (3)
Claim language	en

Short verdict (Claude)

Kurzfazit US20250000006A1

> Text US20250000006A1

Claim chart: Claim 1 (composition)

Result (computed from the feature rows)	literally possible
Model	literally possible
Red team	literally not met
Final	literally possible

FEATURE	CLAIM WORDING (QUOTE)	QUOTE CHECK	PRODUCT	VERDICT	REASONING / CONSTRUCTION / CLARIFICATION
M1	*A lipid nanoparticle comprising an ionizable lipid having a pKa of 6.0 to 6.8*	✔ found verbatim in the claim	46,3 mol-% [P1]	? unclear	Begruendung M1 — Construction: broad = weit; narrow = — Clarification: Messung
M2	*an mRNA*	✔ found verbatim in the claim	46,3 mol-% [P1]	■ met	Begruendung M2 — Construction: broad = weit; narrow =

Red team (patentee view)

Further routes of attack

- indirect infringement (medium): Lieferung

Revised risk of the red team: medium (3). rt

Open questions for counsel

- Gemessene mol-% im Partikel?
- Analytik
- Claim 1 M1: Messung

Design-around ideas (ideas only: technical hypotheses, to be validated technically and legally)

- M2: mol-% unter 40 [narrow margin] - fachlich zu validieren

Exemptions and role (model view)

Experimental-use exemption (§ 11 No. 2 PatG)	open
Regulatory (Bolar) exemption (§ 11 No. 2b PatG)	yes
US safe harbor (35 U.S.C. § 271(e)(1))	n/a
Tipping point	ab Angebot Ausnahmen
Direct infringement by the user	yes
Indirect / contributory infringement	open Rolle

5. Cross-cutting points (Claude; from the individual assessments)

Cross-patent patterns

> Querschnitt.

Timeline (expiry dates are estimates)

PATENT RIGHT	EXPIRY (ESTIMATE)	NOTES
EP1000001B1	2035-06-01	Schätzung
EP1000008B1	2035-06-01	Schätzung
EP1000009B1	2035-06-01	Schätzung
US9000003B2	2035-11-01	Schätzung
EP1000004B1	2035-06-01	Schätzung
EP1000002B1	2035-06-01	Schätzung
US20250000006A1	2035-06-01	Schätzung

Exemptions

> Ausnahmen.

Design-around ideas cross-checked (ideas only)

CHANGE	PATENT RIGHT	COLLIDES WITH	ASSESSMENT
mol-% ändern	EP1000001B1		technische Hypothese

Red team

> Gegenprüfung.

Pending applications

> Anhängig.

6. Patent rights not analysed, and why

no claim text in the corpus (2)

PATENT RIGHT	TITLE	LEGAL STATUS	EXPIRY (ESTIMATE)	REASON
EP1000012B1	TITLE EP1000012B1	granted, in force in DE (unverified, no record; treated as in force)	2035-06-01	no claim text in the corpus (biblio only); check a family member
EP1000011B1	TITLE EP1000011B1	granted, in force in DE (snapshot 2026-08-23)	2035-06-01	no claim text in the corpus (biblio only); check a family member

legal status (expired/lapsed/revoked) (1)

PATENT RIGHT	TITLE	LEGAL STATUS	EXPIRY (ESTIMATE)	REASON
EP1000005B1	TITLE EP1000005B1	expired (snapshot 2026-08-23)	2020-03-01	Legal status: expired (snapshot 2026-08-23); expiry (estimate) 2020-03-01; no claim chart (only at depth "deep")

triage C (different subject) (1)

PATENT RIGHT	TITLE	LEGAL STATUS	EXPIRY (ESTIMATE)	REASON
EP1000007B1	TITLE EP1000007B1	granted, in force in DE (snapshot 2026-08-23)	2035-06-01	Triage C: Grund EP1000007B1

7. Next steps

PRIORITY	STEP	WHY
1	Markush-Recherche	Lücke

Always recommended (method)

- Markush/structure search in CAS STN (REGISTRY + MARPAT) and/or Derwent Markush for the ionizable lipid and the PEG-lipid.

- Current legal status from the registers (EP Register incl. opposition file, DPMAregister, UPC register/opt-out, USPTO Patent Center, PTAB), INPADOC family, SPC/PTE.
- File-wrapper review for the top-rated rights (amendments, statements).
- Analytical data to resolve unclear features (measured mol-% in the particle, size by the method named in the patent, pKa, encapsulation).
- Written opinion of a patent attorney before GMP or commercial commitments.
- Monitoring of pending applications and divisionals/continuations of the affected families and the watched applicants.

Open questions

- Frage 1

Usage

18 calls; 878450 / 26500 bytes in/out

How to read this report

RISK SCALE (1 TO 5; "CANNOT BE ASSESSED" IS NOT A LOW RISK)

●○○○○ very low (1) ●●○○○ low (2) ●●●○○ medium (3) ●●●●○ high (4)

●●●●● very high (5) ? cannot be assessed

MARKS

- ✔ found verbatim in the claim quote found verbatim in the claim text (checked by code)
- ≈ DEVIATING quote found only with an omission or relaxed spelling, to be checked by hand
- ✖ NOT VERIFIED quote not verified; the row is treated as unclear

Verdict "met" means the claim feature is found in the product; it is the adverse reading for the product.

METHOD

Quotes are verified against the claim text, numeric ranges and structure matches are computed in code, and the final risk is the more adverse of the claim chart and the red-team review. The model only translates and explains. Evidence computed by the app only adds flags, questions and risk floors; it never lowers a risk.

Important notice – limits of this FTO pre-screen

1. This is an automated (AI-assisted) pre-screen. It is not legal advice and not a freedom-to-operate opinion, and it does not replace review by a patent attorney.
2. The screened corpus is restricted to the LNP/lipid field (EP/US/WO; defined CPC classes, keywords and applicants). Rights outside this restriction – including utility models, national rights of other countries and supplementary protection certificates – are not covered.
3. Markush/generic claims may cover structures that are not individually indexed in the corpus. No Markush search (e.g. CAS STN MARPAT) was performed. The formula test is only applied to claims whose text is in the corpus; formulae embedded as images may be missing.
4. Legal status (grant, opposition, limitation, lapse, term, PTA/PTE, SPC, UPC opt-out) reflects a data snapshot of 2026-08-23 and may be stale or incomplete. Expiry dates are estimates.
5. Claim texts may be machine translations or OCR text; only the text in the language of proceedings, as granted or maintained, is authoritative.
6. Pending applications: claims may change before grant; divisionals and continuations may be filed. Applications not yet published (up to 18 months from priority) are not covered.
7. The assessment rests on the product/process data and assumptions listed under "Subject of the screen". If they change, the result changes.
8. Claim construction, equivalence and exemptions (experimental use, Bolar, safe harbor) depend on the individual case and current case law; prosecution files were not reviewed.
9. AI models make mistakes (wrong substituent mapping, wrong chain-length counts, overlooked provisos). Verify quotes and feature mappings against the original document before any decision.